

check the compliance status of an S3 bucket using AWS Config

Goal:

Check whether an S3 bucket complies with a specific rule — e.g., **encryption enabled**, **public access blocked**, or **versioning turned on** — using **AWS Config**.

Step-by-Step Guide: AWS Console

◆ **Step 1: Enable AWS Config**

1. Go to **AWS Console** → **AWS Config**
 2. Click “**Set up AWS Config**”
 3. Select:
 - **Record all resources**
 - **Include global resources** (for S3)
 4. Choose or create:
 - **S3 bucket** to store snapshots
 - **SNS topic** (optional) for notifications
 5. Click **Next** → **Confirm**
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◆ **Step 2: Add S3 Compliance Rules**

AWS provides **managed rules**. Example rules:

Rule Name	Purpose
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<code>s3-bucket-server-side-encryption-enabled</code>	Ensures SSE is enabled on all S3 buckets
<code>s3-bucket-public-read-prohibited</code>	Ensures no bucket is publicly readable
<code>s3-bucket-logging-enabled</code>	Ensures access logs are enabled for S3
<code>s3-bucket-versioning-enabled</code>	Ensures versioning is turned on

To add a rule:

1. Go to **AWS Config** → **Rules** → **Add Rule**
2. Search for `s3`
3. Select e.g., `s3-bucket-server-side-encryption-enabled`
4. Provide a name (optional), keep default trigger settings
5. Click “**Add rule**”

◆ Step 3: View Compliance Results

1. Go to **AWS Config** → **Rules**
2. Click on your rule (e.g., `s3-bucket-server-side-encryption-enabled`)
3. See compliance summary:
 -  **Compliant buckets**
 -  **Non-compliant buckets**
 - Reason for non-compliance

You can also view **resource timeline**:

- Go to **AWS Config** → **Resources** → **S3 Bucket**

- Select your bucket → view **changes**, **compliance history**
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Optional: Remediate Automatically

- Use **AWS Systems Manager Automation** documents (SSM runbooks) to auto-fix non-compliant buckets.
 - Example: Enable encryption using an SSM document when a bucket is found non-compliant.
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CLI Option (Advanced)

```
aws configservice describe-compliance-by-resource \  
--resource-type AWS::S3::Bucket \  
--resource-id your-bucket-name
```

Architect/Exam Tips (SAA-C03)

- **S3 is a global service**, but AWS Config is **regional**. Enable **global resource recording**.
 - Use AWS Config **aggregator** for multi-region/multi-account views.
 - AWS Config **only detects drift**, it does **not enforce** changes.
 - Combine with **CloudTrail**, **SNS**, and **Systems Manager** for full compliance automation.
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